

Three completions of the EVQ-COSH theory: a constant-free τ , a LoRA rank floor, and a prior-robustness audit

Derivations against `paper-2027`, App. A.1/A.8/A.10/A.12 and Thm. 3

Abstract

Three derivations built only from objects already in the manuscript. **(I)** The protocol constant $c(\Pi)$ in $\tau = c(\Pi)d_{\text{eff}}/\sqrt{L_{\text{train}}}$ is removed. The softmax-Jacobian window L_{eff}^J replaces L *exactly* (not approximately) in the A.10 utility, and the coefficient $Q_1(L, b)$ admits a closed form $Q_1 = \frac{1}{12}\varphi_*(1 - \varphi_*)(2 - \varphi_*) + O(\log^{-2} b)$ with $\varphi_* = \log(L/x_0)/\log b$ and a universal $x_0 = 2.0743$. The resulting rule has zero free constants and reproduces every deployed τ in the paper to 5–9% (exactly at $L=512$) — with one exception, the $b=256$ fixed-support control, which sits on the formula’s degeneracy locus. $c=1$ turns out to be not a convention but the *ceiling* of the model, attained only on the curve $b = (L/x_0)^{(3+\sqrt{3})/2}$. Feeding in the repository’s existing κ_{att} audit gives $L_{\text{eff}}^J/L_{\text{train}} \approx 4$ — attention locality *lengthens* this window, because the transport direction vanishes on near keys — and the fixed point of the self-consistent equation lands at $\tau \approx 0.5 \times$ the deployed rule, a multiplier no reported sweep covers. **(II)** Theorem 3 is upgraded to a quantitative rank floor: a rank- r Q/K LoRA leaves a subspace of codimension $\leq 2r$ on which the un-repaired transplant survives verbatim, which forces $r \geq d_{\text{rot}}/2$ for exact single-distance repair and $r \geq (1 - \eta) \sum_k s_k/8$ with $s_k = 2(1 - \text{Re} \psi_\mu(\omega_k - \omega'_k))$ for η -approximate repair; the small-shift limit of $\sum_k s_k$ is $\sigma_\mu^2 K W_2^2(\Omega, \Omega')$. The deployed $r=64$ sits exactly at the exact-repair floor and $\approx 3 \times$ above the 90% floor. The rank analysis does *not* separate allocation from support — it separates *slow-only* from *full-band*. **(III)** App. A.1 generalises verbatim once $(\sin t/t, (1 - \cos t)/t)$ is read as $(\text{Re} \psi_\mu, \text{Im} \psi_\mu)$. The $r_2=2.00$ slow-collapse figure is prior-robust. But under a Zipf prior with tail index $\alpha \geq 1.5$ the $r_2(\tau)$ curve acquires an interior maximum at $\tau \approx 3.5$ –6: high-frequency overload is real. It never reverses the Geo/EVQ-COSH ordering, but it erases the margin — and it agrees with Part I that the deployed τ is on the high side. The ceiling is diagnosed exactly: the A.10 utility is diagonal in the channel index, so it cannot see the overload term at all, and the agreement is a coincidence of sign rather than a derivation.

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1 Part I: a self-consistent τ

1.1 Setting and standing assumptions

Write $\theta = \tau^2$. From App. A.12, the EVQ-COSH density perturbation is $\rho_\tau(\phi) = 1 + \theta a(\phi) + O(\theta^2)$ with $a(\phi) = \frac{(1-\phi)^2}{2} - \frac{1}{6}$, so $\int_0^1 a = 0$ (mass conservation). From App. A.10–A.11 the two competing terms are the Pearson- χ^2 channel-load stiffness and the post-softmax phase-variance utility,

$$S_{\chi^2}(\tau) = \frac{\tilde{S}(\tau)}{d_{\text{eff}}}, \quad \tilde{S}(\tau) = \frac{\sinh \tau \arctan(\sinh \tau)}{\tau^2} - 1 = \frac{\tau^4}{45} + O(\tau^6), \quad U(\tau) = \frac{d_{\text{eff}}}{L} \int_0^1 \rho_\tau(\phi) q(Lb^{-\phi}) d\phi, \quad (1)$$

with $q(x) = \text{Var}_{t \sim U[0,1]}[\cos(xt)] = \frac{1}{2} + \frac{\sin 2x}{4x} - (\frac{\sin x}{x})^2$, $q(x) = x^4/45 + O(x^6)$ and $q(\infty) = \frac{1}{2}$. Expanding ρ_τ recovers the paper's $U = (d_{\text{eff}}/L)[Q_0 + \tau^2 Q_1 + O(\tau^4)]$ with $Q_0 = \int_0^1 q(Lb^{-\phi}) d\phi$, $Q_1 = \int_0^1 a(\phi) q(Lb^{-\phi}) d\phi$, and stationarity of $\mathcal{F} = \frac{1}{2} S_{\chi^2} - \lambda U$ gives $\tau_*^2 = 45\lambda Q_1 d_{\text{eff}}^2/L$, i.e. $c(\Pi) = \sqrt{45\lambda Q_1(L, b)}$.

Two structural facts about (1) drive everything below. The d_{eff} in U is a *channel count* ($K = d_{\text{eff}}/2$ rotary pairs, each contributing q); the $1/L$ is a *key count* (the number of positions the budget must resolve), inherited from the diffuse baseline $p_0 = 1/L$. These are two different objects and only coincide when attention is uniform on the window.

Assumption 1.1 (standing). (A1) $\lambda > 0$ is a fixed exchange rate between load and utility. (A2) $d_{\text{eff}} = d_{\text{rot}}$ (MHA: $= d_{\text{head}}$; compressed-RoPE MLA: $= d_{\text{rope}}$), matching App. A.13. (A3) Frozen-Jacobian: the trained attention distributions p_i and the logit transport directions g_i of App. A.12 are evaluated on the $\tau=0$ substrate and held fixed while τ varies. (A4) $\mathbb{E}_{\ell, h, i} \|P_i g_i\|_2^2 > 0$, i.e. the transport direction is not p_i -a.s. constant.

1.2 Why the obvious route to a constant-free τ fails

The natural first attempt is to delete λ by replacing the soft penalty with a trust region on the attention displacement itself, using the KL identity of App. A.12. It cannot work, for a reason worth recording.

Lemma 1.2 (Order degeneracy of the two second-order costs). *Under Assumption 1.1, the EVQ-COSH warp displaces the attention distribution by*

$$D_{\text{KL}}(p \parallel p_\tau) = \frac{1}{2}\theta^2 \mathbb{E}[g^\top J_{\text{sm}}(p)g] + O(\theta^3) = \frac{\tau^4}{2} \cdot \frac{\mathbb{E}\|Pg\|_2^2}{L_{\text{eff}}^J} + O(\tau^6),$$

which is $O(\tau^4)$ — the same order as the channel-load stiffness $S_{\chi^2}(\tau) = \tau^4/(45d_{\text{eff}}) + O(\tau^6)$. Their ratio is τ -independent at leading order. Consequently no criterion built only from these two objects (their ratio, or a trust region $D_{\text{KL}} \leq \delta$) can select a finite τ with the observed scaling: the trust region gives $\tau \propto \delta^{1/4}(L_{\text{eff}}^J)^{1/4}$, wrong in both exponent and sign.

Proof. $g = \partial_\theta z|_{\tau=0}$ by definition (App. A.12, Eq. (g-closed-form)), so the schedule change is $\delta z = \theta g + O(\theta^2)$ and $p_\theta = \text{softmax}(z + \theta g)$. The KL expansion is the definition of the softmax Fisher metric; the second equality is $\mathbb{E}[g^\top J_{\text{sm}}g] = \kappa_{\text{att}} \mathbb{E}\|Pg\|^2$, i.e. the ratio-of-means definition (2) below. S_{χ^2} is App. A.11. The ratio of the two is $45d_{\text{eff}} \mathbb{E}\|Pg\|^2/(2L_{\text{eff}}^J)$, free of τ . Solving $\frac{1}{2}\tau^4 \mathbb{E}\|Pg\|^2/L_{\text{eff}}^J = \delta$ gives the stated scaling. $\square$

So a τ^2 -order utility is structurally necessary, and A.10 supplies exactly that. The route to a constant-free rule is therefore not to remove λ but to *derive the entire (L, b) dependence of c* , leaving λ as one universal number fixed once by a normalisation convention.

1.3 L_{eff}^J replaces $1/L$ exactly

Recall the primary definition of App. A.12,

$$\frac{1}{L_{\text{eff}}^J} = \kappa_{\text{att}} = \frac{\mathbb{E}_{\ell, h, i}[g_i^\top J_{\text{sm}}(p_i)g_i]}{\mathbb{E}_{\ell, h, i}[\|P_i g_i\|_2^2]}, \quad J_{\text{sm}}(p) = \text{diag}(p) - pp^\top. \quad (2)$$

Proposition 1.3 (Exact reduction, not an approximation). *Let p be uniform on a key set of size L and let $Pg \neq 0$. Then $\kappa_{\text{att}} = 1/L$ exactly, for every g , with no expansion and no genericity assumption.*

Proof. $g^\top J_{\text{sm}}(p)g = \text{Var}_{j \sim p}[g(j)] = \frac{1}{L} \sum_j (g_j - \bar{g})^2$ while $\|Pg\|_2^2 = \sum_j (g_j - \bar{g})^2$. The ratio is $1/L$ identically. (Verified numerically to machine precision at $L = 16, 256, 4096$.) $\square$

This is the licence for the substitution. The $1/L$ in U counts keys; (2) is the softmax-weighted count of keys, and *it agrees with $1/L$ on the nose* when attention is diffuse. Two distinct lengths must now be distinguished, however:

- L_{eff}^J — the **effective key count** (normalisation), entering as κ_{att} ;
- L_{rng} — the **effective phase range**, entering inside $q(\omega L_{\text{rng}})$, i.e. the largest relative distance whose phase must be resolved.

Under $p = \text{Unif}$ these coincide with L_{train} . Under a heavy-tailed attention profile they do not: L_{eff}^J measures concentration, L_{rng} measures reach. This distinction returns in §3.5.

Table 1: Accuracy of the closed form (4) against numerical quadrature of $Q_1 = \int_0^1 a(\phi)q(Lb^{-\phi})d\phi$. Predicted error order $\log^{-2} b$ is 0.58% at $b=5 \times 10^5$, matching the observed 0.4–0.5%.

configuration	L	b	φ_*	Q_1 (exact)	Q_1 (4)	err.	Q_0 exact/closed
50M diagnostic	512	5×10^5	0.420	0.03192	0.03207	+0.5%	0.2099 / 0.2099
151.9M fixed-support	256	256	0.868	0.01069	0.01078	+0.8%	0.4333 / 0.4342
432M MLA	8192	5×10^5	0.631	0.02646	0.02656	+0.4%	0.3155 / 0.3155
750M / OLMo-2 1.485B	4096	5×10^5	0.578	0.02878	0.02889	+0.4%	0.2891 / 0.2891
LLaMA-3-8B	8192	5×10^5	0.631	0.02646	0.02656	+0.4%	0.3155 / 0.3155

1.4 Closed form for $Q_1(L, b)$

q is a soft step: $q(x) = x^4/45$ for $x \ll 1$ and $q \rightarrow \frac{1}{2}$ for $x \gg 1$, with the transition at $\omega L \sim 1$ — precisely the collapse threshold of Prop. 2 (low-frequency collapse). In the substitution $x = Lb^{-\phi}$ the transition occupies a ϕ -window of width $O(1/\log b)$, so for large b the step is sharp on the ϕ axis. This makes Q_0, Q_1 elementary.

Proposition 1.4 (Closed-form utility coefficients). *Let $x_0 > 0$ be the unique root of the log-moment balance*

$$\int_0^\infty [q(x) - \tfrac{1}{2}\mathbf{1}\{x > x_0\}] \frac{dx}{x} = 0, \quad x_0 = 2.0743\dots \quad (3)$$

and set $\varphi_*(L, b) = \min\{1, \log(L/x_0)/\log b\}$, $g(\varphi) = \varphi(1-\varphi)(2-\varphi)$. Then as $b \rightarrow \infty$ with φ_* fixed in $(0, 1)$,

$$Q_0(L, b) = \tfrac{1}{2}\varphi_* + O(\log^{-2} b), \quad \boxed{Q_1(L, b) = \tfrac{1}{12}g(\varphi_*) + O(\log^{-2} b)}. \quad (4)$$

Proof. Substituting $x = Lb^{-\phi}$ gives $d\phi = -d\log x/\log b$ and $Q_1 = \frac{1}{\log b} \int_{L/b}^L a\left(\frac{\log(L/x)}{\log b}\right) q(x) d\log x$. Because $a \in C^\infty$ varies on the ϕ -scale 1 while q varies on the ϕ -scale $1/\log b$, Taylor-expanding a about φ_* inside the transition window gives

$$Q_1 = \tfrac{1}{2} \int_0^{\varphi_*} a(\phi) d\phi + \underbrace{\frac{a(\varphi_*)}{\log b} \int_0^\infty [q - \tfrac{1}{2}\mathbf{1}\{x > x_0\}] \frac{dx}{x}}_{=0 \text{ by (3)}} + O(\log^{-2} b).$$

The choice (3) annihilates the $O(1/\log b)$ term, which is why the remainder is $O(\log^{-2} b)$. Finally $\int_0^u a = \frac{1}{6}u(1-u)(2-u) = \frac{1}{6}g(u)$ by direct integration, giving (4); Q_0 is the same computation with $a \equiv 1$. Uniqueness of x_0 follows from strict monotonicity of the left side of (3) in x_0 . $\square$

Remark 1.5 (Interpretation of x_0). $x_0 = 2.074$ is the utility-weighted resolution threshold: a channel counts as “resolved” when $\omega L \gtrsim 2$, i.e. when its wavelength is below $\approx 3L$. The paper’s informal $\omega L \leq 1$ cut used in the $r_2=2.00$ example is the same threshold read off the *geometry*; x_0 is its *utility*-weighted counterpart. Note also the exact identity $\int_0^1 a = 0$: if every channel lies on one side of the threshold ($\varphi_* \in \{0, 1\}$) then $Q_1 = 0$ identically. **The first-order utility of EVQ-Cosh comes entirely from the existence of a resolution boundary inside the band,** which is Prop. 2 (collapse) re-derived from the operating-point side.

1.5 The self-consistent equation

Theorem 1.6 (Self-consistent operating point). *Under Assumption 1.1, replace $1/L$ in U by $\kappa_{\text{att}} = 1/L_{\text{eff}}^J$ and L inside q by L_{rng} , and use the exact (non-truncated) $\tilde{S}$ of (1). Let $W(\tau) = \int_0^1 \rho_\tau(\phi) q(L_{\text{rng}} b^{-\phi}) d\phi$. Then every interior stationary point of $\mathcal{F}(\tau) = \frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau)$ satisfies the **self-consistency equation***

$$\boxed{\Psi(\tau_*) = \frac{2\lambda d_{\text{eff}}^2}{L_{\text{eff}}^J}, \quad \Psi(\tau) := \frac{\tilde{S}'(\tau)}{W'(\tau)}. \quad (5)}$$

Ψ is strictly increasing on $(0, \infty)$ with $\Psi(0^+) = 0$ and $\Psi(\infty) = \infty$ (verified numerically over $\tau \in [0.2, 8]$ for all paper configurations), so τ_* exists and is unique. Its small- τ expansion, $\Psi(\tau) = 2\tau^2/(45Q_1) + O(\tau^4)$, reproduces

$$\tau_*^2 = \frac{45\lambda Q_1(L_{\text{rng}}, b) d_{\text{eff}}^2}{L_{\text{eff}}^J}, \quad \text{i.e.} \quad \tau_* = \sqrt{45\lambda Q_1} \frac{d_{\text{eff}}}{\sqrt{L_{\text{eff}}^J}}. \quad (6)$$

This equation is self-consistent in the strict sense: L_{eff}^J on the right depends, through the trained p_i , on the substrate built with τ_* ; (5) is a fixed point in τ , solved in practice by one measure-rebuild iteration.

Proof. $\partial_\tau \mathcal{F} = 0$ reads $\frac{1}{2d_{\text{eff}}} \tilde{S}'(\tau) = \lambda \frac{d_{\text{eff}}}{L_{\text{eff}}^J} W'(\tau)$; rearranging gives (5). The expansion uses $\tilde{S}' = 4\tau^3/45 + O(\tau^5)$ and $W' = 2\tau Q_1 + O(\tau^3)$ from $\rho_\tau = 1 + \tau^2 a + O(\tau^4)$. $\square$

1.6 Removing the constant

(6) still carries λ . Combining it with Prop. 1.4,

$$c(\Pi) = \sqrt{45\lambda Q_1} = \sqrt{\frac{45\lambda}{12} g(\varphi_*)}, \quad g(\varphi) = \varphi(1 - \varphi)(2 - \varphi). \quad (7)$$

The function g on $[0, 1]$ is maximised at $\varphi_{\text{max}} = 1 - 1/\sqrt{3}$ with $g_{\text{max}} = g(\varphi_{\text{max}}) = \frac{2}{3\sqrt{3}} = \frac{2\sqrt{3}}{9} = 0.38490$ (exact). This gives $c(\Pi)$ a hard ceiling, and the ceiling is what fixes λ .

Corollary 1.7 (Constant-free operating rule). *Adopt the normalisation “the rule is exact at the best-conditioned configuration”, i.e. $c = 1$ when $\varphi_* = \varphi_{\text{max}}$. This forces $\lambda = \frac{12}{45g_{\text{max}}} = \frac{4}{15g_{\text{max}}} = 0.6928$ and yields*

$$\boxed{\tau_* = \frac{d_{\text{eff}}}{\sqrt{L_{\text{eff}}^J}} \sqrt{\frac{g(\varphi_*)}{g_{\text{max}}}} = \frac{d_{\text{eff}}}{\sqrt{L_{\text{eff}}^J}} \sqrt{\frac{3\sqrt{3}}{2} \varphi_*(1 - \varphi_*)(2 - \varphi_*)}, \quad \varphi_* = \min \left\{ 1, \frac{\log(L_{\text{rng}}/x_0)}{\log b} \right\}. \quad (8)}$$

No protocol constant remains. In particular $c(\Pi) \leq 1$ always, with equality **iff** $b = (L_{\text{rng}}/x_0)^{1/\varphi_{\text{max}}} = (L_{\text{rng}}/x_0)^{(3+\sqrt{3})/2}$.

This is the substantive change to the paper’s claim. $c(\Pi)$ is *not* a free, protocol-dependent number. Its entire (L, b) dependence is closed-form; only a single universal normalisation remains, and $c=1$ is the model’s maximum rather than a typical value.

Table 2: Constant-free rule (8) against the deployed τ , evaluated at $L_{\text{eff}}^J = L_{\text{rng}} = L_{\text{train}}$ (uniform-attention hypothesis).

configuration	d_{eff}	L_{train}	b	φ_*	$c(\text{II})$	τ predicted	τ deployed
50M diagnostic	64	512	5×10^5	0.420	1.000	2.828	2.830
432M MLA ($d_{\text{rot}}=32$)	32	8192	5×10^5	0.631	0.910	0.322	0.354
750M continuation	64	4096	5×10^5	0.578	0.949	0.949	1.000
OLMo-2 1.485B	128	4096	5×10^5	0.578	0.949	1.898	2.000
LLaMA-3-8B	128	8192	5×10^5	0.631	0.910	1.287	1.414
151.9M fixed-support	64	256	256	0.868	0.580	2.318	4.000

Table 3: Exact solution of (5) versus the first-order rule, with λ calibrated once at $(L, b) = (4096, 5 \times 10^5)$. The last column shows how badly the quartic truncation of the stiffness alone fails at the deployed τ — and how little it matters.

configuration	τ deployed	τ first-order	τ exact	$\tilde{S}_{\text{exact}}/(\tau^4/45)$
50M diagnostic	2.830	2.979	2.991	0.373
432M MLA	0.354	0.339	0.343	0.966
750M continuation	1.000	1.000	1.082	0.785
OLMo-2 1.485B	2.000	2.000	2.158	0.507
LLaMA-3-8B	1.414	1.356	1.494	0.656
151.9M fixed-support	4.000	2.438	2.428	0.284

1.7 Is $\tau = d_{\text{eff}}/\sqrt{L_{\text{train}}}$ a strict first-order approximation?

Theorem 1.8 (Split verdict). *Let attention be uniform on the training window, $p_i = 1/L_{\text{train}}$.*

- (a) **Confirmed, and stronger than first order.** *The length dependence $\tau_* \propto d_{\text{eff}}/\sqrt{L_{\text{train}}}$ is exact, not a first-order approximation: by Prop. 1.3, $L_{\text{eff}}^J = L_{\text{train}}$ identically, and $L_{\text{rng}} = L_{\text{train}}$ by definition of the window. The only approximation in the paper’s rule is the small- τ truncation of (5), and this is numerically benign: solving (5) exactly (no Taylor series in either $\tilde{S}$ or W) shifts τ_* by +6% to +8% at the paper’s operating points, even though the quartic truncation of $\tilde{S}$ alone is off by a factor 0.28–0.79 there. The errors in $\tilde{S}'$ and W' cancel to leading order in the ratio Ψ .*
- (b) **Refuted.** *$c = 1$ does not follow. By (7), $c(\text{II})$ is a nonconstant function of (L, b) through φ_* , and requiring $c \equiv 1$ across the paper’s configurations forces λ to vary by a factor 2.99. Excluding the $b=256$ control, the required λ varies only over $[0.696, 0.840]$ (a factor 1.21), so $c \approx 1$ is a good approximation on the $b=5 \times 10^5$ family — but it is an approximation to the ceiling, not a derived identity.*

1.8 Where the approximation fails completely

Proposition 1.9 (Failure boundaries). *The rule $\tau = c d_{\text{eff}}/\sqrt{L_{\text{train}}}$ degenerates in four distinct ways.*

- (F1) **Band degeneracy**, $\varphi_* \rightarrow 0$ **or** 1. $Q_1 \rightarrow 0$ and $\tau_* \rightarrow 0$: *if every channel is on one side of the resolution threshold there is no first-order gain from reallocation, because $\int_0^1 a = 0$. The*

boundary $\varphi_* = 1$ is $b = x_0 L \approx 2.07L$. The paper’s own fixed-support control sits on this locus: $b=256$, $L=256$ gives $\varphi_* = 0.868$ and $c = 0.58$, so the deployed $\tau=4$ overshoots the model’s optimum (2.32–2.43) by 1.6–1.7 $\times$. That the control nonetheless favours EVQ-COSH at $\tau=4$ is therefore evidence against the operating-point model at that configuration, not for it — and it is the cleanest place in the paper to test whether the mechanism is the modelled one.

- (F2) **Rayleigh-quotient degeneracy.** If $P_i g_i \rightarrow 0$ — attention on a single key, or on keys at a single relative distance — then (2) is 0/0 and L_{eff}^J is undefined. Assumption (A4) is not cosmetic: $g_i(j) \propto r_{ij} \times (\cdot)$ vanishes at $r_{ij} = 0$, so a head that attends only to a narrow band of distances has a nearly constant g_i and an ill-conditioned L_{eff}^J . The estimator must be reported with $\mathbb{E}\|P_i g_i\|^2$ alongside it.
- (F3) **Large- τ truncation.** At $\tau=4$ the exact stiffness is 0.28 times its quartic. This is harmless for τ_* (Table 3) but not for any statement about the value of $\mathcal{F}$, e.g. the waterbed scale $\mathcal{W}(\rho_{\tau_*}) = O(d_{\text{head}}^4 / (L_{\text{eff}}^J)^2)$ in App. A.6, which inherits the full 0.28 error.
- (F4) **Decoupling of L_{eff}^J from L_{rng} .** The substitution repairs the key-count normalisation only. When attention is heavy-tailed the two lengths diverge and the formula’s prescription conflicts with the geometry; see §3.5.

1.9 The measurement already exists, and it points the other way

(8) makes L_{eff}^J falsifiable, and the repository already contains the falsification: `research/KAPPA_ATTENTION_MEASURE_AUDIT_20260820.md` (implementation `scripts/analysis/attention_fisher_50m_probe.py`) measures κ_{att} on the four cells of the frozen 50M co-adaptation probe at $L=512$, with the direction g computed once on the Geo cell and held fixed — exactly Assumption (A3). Reading $L_{\text{eff}}^J = 1/\kappa_{\text{att}}$ off the first-order column:

weights	table	PPL	κ_{att}	L_{eff}^J	$L_{\text{eff}}^J / L_{\text{train}}$
Geo	Geo	7.141	4.648×10^{-4}	2152	4.20
Geo	EVQ-COSH	76.20	2.531×10^{-3}	395	0.77
EVQ-COSH	Geo	23.05	1.144×10^{-3}	874	1.71
EVQ-COSH	EVQ-COSH	7.160	5.354×10^{-4}	1868	3.65

Two facts follow, one of which reverses an intuition that would otherwise have been natural here.

Proposition 1.10 (The transport direction vanishes where attention concentrates). *By App. A.12, $g_i(j) = \frac{(\log b) r_{ij}}{s_{\text{att}}} \sum_k \omega_k^{(0)} h(u_k)[\cdot]$, so $g_i(j) = O(r_{ij})$ as $r_{ij} \rightarrow 0$ and $g_i(i) = 0$ exactly. Trained attention places most of its mass at small $|r_{ij}|$, i.e. precisely where $|g_i|$ is smallest. Hence $\text{Var}_{p_i}[g_i] < \frac{1}{n} \|P_i g_i\|^2$ and $\kappa_{\text{att}} < 1/n$, i.e. $L_{\text{eff}}^J > L_{\text{train}}$: **attention locality makes the softmax-Jacobian window longer, not shorter.** The measurement confirms this: against a uniform-attention reference of $\kappa_0 \in [1/512, 1/269]$ for this probe’s causal query geometry (query positions 63, 127, 255, 383, 511), the two self-consistent cells measure κ_{att} a factor 4.2–8 smaller.*

This inverts the naive reading of (8). One is tempted to argue “real attention is concentrated, so $L_{\text{eff}}^J \ll L_{\text{train}}$, so the deployed τ is too small”. The opposite is true.

Corollary 1.11 (The 50M fixed point). *At the 50M configuration $\varphi_* = 0.420 \approx \varphi_{\max}$, so $c(\Pi) = 1.000$ exactly and (8) reduces to $\tau_* = d_{\text{eff}}/\sqrt{L_{\text{eff}}^J}$. Substituting the two self-consistent measurements,*

$$\tau_* = \frac{64}{\sqrt{2152}} = 1.380 \quad (\text{Geo substrate}), \quad \tau_* = \frac{64}{\sqrt{1868}} = 1.481 \quad (\text{EVQ-COSH } \tau=2.83 \text{ substrate}).$$

The map $\tau \mapsto L_{\text{eff}}^J(\tau) \mapsto d_{\text{eff}}/\sqrt{L_{\text{eff}}^J}$ is therefore strongly contractive here ($[0, 2.83] \mapsto [1.38, 1.48]$), with fixed point

$$\tau^{\text{sc}} \approx 1.43 = 0.51 \times \text{the deployed rule.}$$

A concrete, untested prediction. The manuscript’s multiplier sweeps evaluate $0.75\times$, $1.00\times$, $1.25\times$ and $1.50\times$ the rule. **The self-consistent formula predicts the optimum at $\approx 0.5\times$, which no reported sweep covers.** If the $L_{\text{eff}}^J/L_{\text{train}} \approx 4$ ratio is not special to 50M, the same halving applies everywhere: $\tau \approx 1.0$ for OLMo-2 1.485B rather than 2, ≈ 0.7 for LLaMA-3-8B rather than 1.414. Adding a $0.5\times$ arm to the existing 99-run staged study is the single cheapest decisive test of this note.

Honest carry-over of the audit’s own verdict. The audit’s preregistered promotion test was whether κ_{att} *ranks trained PPL across the four cells*. The first-order direction does ($\rho=+1.0$); the finite-swap direction does not ($\rho=-0.2$); the preregistered rule therefore selected Branch C and κ_{att} was not promoted. That verdict concerns κ_{att} *as a PPL predictor*. The use here is different and weaker: κ_{att} enters (5) only as the *normalisation* of a diagonal utility, replacing an assumed $p_0=1/L$ by a measured one, and Prop. 1.3 shows the replacement is exact at the point where the two agree. The disagreement between the two directions is nevertheless a real caveat: it bounds how much weight the fixed point of Cor. 1.11 can carry until a $0.5\times$ training arm exists.

2 Part II: a LoRA rank floor from the transplant obstruction

2.1 Setup

Work in head coordinates on $\mathbb{R}^{d_{\text{rot}}}$, $K = d_{\text{rot}}/2$. Let $R_{\Omega}(\Delta) = \bigoplus_{k=1}^K \text{Rot}(\omega_k \Delta)$ and $D(\Delta) := R_{\Omega}(\Delta) - R_{\Omega'}(\Delta)$, $\delta_k := \omega_k - \omega'_k$ under the monotone (sorted) pairing, which is the W_p -optimal coupling of the two frequency multisets in one dimension. A rank- r LoRA gives effective maps $A = W_Q + \Delta Q$, $B = W_K + \Delta K$ with $\text{rank } \Delta Q, \text{rank } \Delta K \leq r$; take $W_Q = W_K = I$ in head coordinates (the general case only adds fixed invertible factors, which do not change ranks).

2.2 The residual is low rank, uniformly in Δ

Lemma 2.1 (Rank- $2r$ residual). *For every Δ , $\text{rank}(A^{\top} R_{\Omega'}(\Delta) B - R_{\Omega'}(\Delta)) \leq 2r$.*

Proof. $A^{\top} R B - R = \Delta Q^{\top} R B + R \Delta K$, a sum of two matrices of rank $\leq r$. (Verified numerically: equality holds generically.) $\square$

Theorem 2.2 (Frozen subspace). *Let $\mathcal{N} = \ker \Delta Q \cap \ker \Delta K$, so $\dim \mathcal{N} \geq d_{\text{rot}} - 2r$, and let P be the orthogonal projector onto $\mathcal{N}$. Then for every Δ ,*

$$P \left(A^{\top} R_{\Omega'}(\Delta) B \right) P = P R_{\Omega'}(\Delta) P, \quad (9)$$

that is, *on a subspace of codimension at most $2r$ the adapted head reproduces the un-repaired transplanted table verbatim*. Consequently, for every Δ ,

$$\left\| A^\top R_{\Omega'}(\Delta)B - R_\Omega(\Delta) \right\|_F \geq \|P D(\Delta) P\|_F. \quad (10)$$

Proof. For $v, w \in \mathcal{N}$: $v^\top A^\top R_{\Omega'} B w = (Av)^\top R_{\Omega'}(Bw) = v^\top R_{\Omega'} w$ since $Av = v$, $Bw = w$. This is (9); subtracting $PR_\Omega P$ and using $\|M\|_F \geq \|PMP\|_F$ gives (10). (Verified numerically to 10^{-15} .) $\square$

This is the precise sense in which “Theorem 3’s obstruction remains in the principal subspace”. It is not a metaphor: the obstruction survives *exactly*, on an explicitly identified subspace whose codimension is the LoRA budget.

2.3 The mismatch spectrum

Lemma 2.3 (Block spectrum). *$D(\Delta)$ is block diagonal with k -th block $\text{Rot}(\omega_k \Delta) - \text{Rot}(\omega'_k \Delta) = 2 \sin(\delta_k \Delta/2) \cdot O_k(\Delta)$ for an orthogonal O_k . Hence $\sigma(D(\Delta)) = \{2|\sin(\delta_k \Delta/2)|\}_{k \leq K}$, each with multiplicity two, and*

$$\|D(\Delta)\|_F^2 = 2 \sum_{k=1}^K 4 \sin^2(\delta_k \Delta/2), \quad \mathbb{E}_{\Delta \sim \mu} \|D\|_F^2 = 2 \sum_k s_k, \quad s_k := 2(1 - \text{Re } \psi_\mu(\delta_k)) \in [0, 4], \quad (11)$$

where ψ_μ is the characteristic function of the evaluation distribution of relative distances. (Verified numerically to 10^{-10} .)

Corollary 2.4 (Exact-repair floor). *If $A^\top R_{\Omega'}(\Delta)B = R_\Omega(\Delta)$ at a single distance Δ with $\delta_k \Delta \notin 2\pi\mathbb{Z}$ for all k , then Lemma 2.1 forces $\text{rank } D(\Delta) \leq 2r$, and $\text{rank } D(\Delta) = 2K$, so*

$$\boxed{r \geq K = d_{\text{rot}}/2.} \quad (12)$$

By Theorem 3 no finite r achieves this simultaneously for all Δ in an interval. Since a global LoRA factors every head’s update through one shared r -dimensional input subspace, the bound must hold for every head at once.

For $d_{\text{head}} = 128$ (OLMo-2 1.485B, LLaMA-3-8B) this reads $r \geq 64$, and the paper deploys $r = 64$: the empirical rank sits exactly on the exact-repair floor, with zero slack.

2.4 Approximate repair: the quantitative bound

Theorem 2.5 (Rank floor for η -approximate repair). *Let μ be the evaluation distribution of relative distances and s_k as in (11). Then for any rank- r Q/K LoRA,*

$$\mathbb{E}_{\Delta \sim \mu} \|A^\top R_{\Omega'}(\Delta)B - R_\Omega(\Delta)\|_F^2 \geq 2 \sum_k s_k - 16r. \quad (13)$$

Hence leaving relative residual energy at most η requires

$$\boxed{r \geq \frac{(1-\eta)}{8} \sum_{k=1}^K s_k = \frac{1-\eta}{4} \left(K - \sum_k \text{Re } \psi_\mu(\delta_k) \right).} \quad (14)$$

A sharper (non-uniform) version replaces the crude $\max_k$ step: with $s_{(1)} \geq \dots \geq s_{(K)}$, $r \geq \frac{1}{2} \min\{m : \sum_{k \leq m} s_{(k)} \geq (1-\eta) \sum_k s_k\}$ whenever the ordering of $|\sin(\delta_k \Delta/2)|$ is Δ -independent (i.e. $\max_k |\delta_k| \cdot \sup \text{supp } \mu \leq \pi$).

Table 4: Mismatch energy and rank floors for $K=64$, $b=5 \times 10^5$, evaluation prior $\text{Unif}[1, 8192]$. The last row uses $\mu = \text{Unif}[1, 256]$ instead. $r(f)$ is the smallest r whose $2r$ worst-matched channels carry a fraction f of $\sum_k s_k$ (the sharp bound of Theorem 2.5).

intervention $\Omega \rightarrow \Omega'$	$\mathbb{E}_\mu \ D\ _F^2$	$r(50\%)$	$r(90\%)$	$r(99\%)$	rigorous (14), $\eta=0$
Native $\rightarrow$ EVQ-COSH($\tau=1.414$)	181.8	12	21	24	11.4
Native $\rightarrow$ EVQ-COSH($\tau=4$)	238.5	15	27	30	14.9
support only: $b : 5 \times 10^5 \rightarrow 5 \times 10^6$	160.4	10	18	22	10.0
support only: PI, all $\omega/4$	164.2	10	19	22	10.3
YaRN-style: only $\omega L < 1$ rescaled	1.07	1	3	6	0.07
151.9M control: FMRoPE $\rightarrow$ anchored EVQ-COSH($\tau=4$), $K=32$, $\mu = \text{Unif}[1, 256]$	119.3	8	14	15	7.5

Proof. By Theorem 2.2 it suffices to bound $\|PD(\Delta)P\|_F$ from below with $\text{codim } \mathcal{N} \leq 2r$. Write $PDP = D - (I - P)D - PD(I - P)$; the two subtracted terms have rank $\leq 2r$ each, so PDP differs from D by a matrix of rank $\leq 4r$ and Weyl's inequality for singular values gives $\sigma_j(PDP) \geq \sigma_{j+4r}(D)$ (verified numerically). Summing squares, $\|PDP\|_F^2 \geq \sum_{j>4r} \sigma_j(D)^2$. By Lemma 2.3 the discarded $4r$ singular values correspond to at most $2r$ channels, each contributing at most $2 \cdot 4 \sin^2(\cdot) \leq 8$; hence $\|PDP\|_F^2 \geq \|D\|_F^2 - 16r$ pointwise in Δ . Taking $\mathbb{E}_\mu$ gives (13), and η -repair requires $2 \sum s_k - 16r \leq \eta \cdot 2 \sum s_k$. The sharper version follows by keeping the sorted tail instead of the crude max. $\square$

2.5 The Wasserstein form

The user-facing question — how does the spectral discrepancy between Ω and Ω' control $r_{\min}$ — is answered by the two regimes of $s_k = 2(1 - \text{Re } \psi_\mu(\delta_k))$.

Corollary 2.6 (Small-shift regime: W_2 controls the floor). *Let $\sigma_\mu^2 = \mathbb{E}_\mu[\Delta^2] < \infty$ and suppose $|\delta_k| \sigma_\mu \ll 1$ for all k . Then $s_k = \delta_k^2 \sigma_\mu^2 + O(\delta_k^4 \mathbb{E} \Delta^4)$ and, since the sorted coupling is W_p -optimal in $\mathbb{R}$,*

$$\sum_k s_k = \sigma_\mu^2 \sum_k \delta_k^2 + O(\cdot) = \sigma_\mu^2 K W_2^2(\Omega, \Omega') + O(\cdot), \quad \text{so} \quad r_{\min} \geq \frac{(1 - \eta) \sigma_\mu^2 K W_2^2(\Omega, \Omega')}{8}. \quad (15)$$

Corollary 2.7 (Saturated regime: a decorrelated-channel count). *If $|\delta_k| \sigma_\mu \gtrsim 1$ then $s_k \rightarrow 2$ and each such channel is fully decorrelated by the transplant. With $K_{\text{sat}} = \#\{k : \text{Re } \psi_\mu(\delta_k) \approx 0\}$, $r_{\min} \geq (1 - \eta) K_{\text{sat}}/4$. The crossover between the regimes is governed by W_∞ : saturation begins once $W_\infty(\Omega, \Omega') \gtrsim 1/\sigma_\mu$.*

So the precise answer is: W_2^2 controls the floor in the perturbative regime, W_∞ controls where that description stops, and beyond it the floor is a count of decorrelated channels. $\sum_k s_k = 2K - 2 \sum_k \text{Re } \psi_\mu(\delta_k)$ is the object that interpolates, and it is computable in closed form for any μ from Prop. 3.1 below.

2.6 Numerical evaluation

Three readings, one of which corrects an attractive but false expectation.

(1) **Slow-only range operators are rank-cheap, and this is Prop. 2 cashing out.** The YARN-style intervention — rescale only the channels with $\omega L < 1$, leaving the fast channels alone — has $170\times$ less mismatch energy and is 90%-repairable at $r = 3$, versus $r \approx 20$ for a full-band

change. This is not a coincidence: those channels are exactly the ones the paper shows have block-whitened $r_2 = 2.00$. **The rank needed to repair an intervention is the effective rank of the subspace the intervention acts on**, so an intervention confined to a two-dimensional positional subspace is absorbed by a rank-2–3 update. This is a new corollary of the paper’s own Prop. 2 and it explains, without any new assumption, why range-transport methods survive with near-zero adaptation while table changes do not.

(2) The rank analysis does *not* separate allocation from support. It would be convenient if EVQ-COSH (allocation) were rank-expensive and a base change (support) were rank-cheap; that would give the “third axis” a rank-theoretic proof. It is false: $r(90\%)$ is 21 for EVQ-COSH and 18–19 for a base change or PI, because all three move every channel. The seam the rank analysis actually cuts along is *slow-only vs. full-band*. This should be stated rather than elided — claiming otherwise is checkable in ten lines of NumPy.

(3) A falsifiable prediction. The floors say the knee is at $r \approx 20$ –27 for the 8B protocol, and that $r \leq 8$ recovers only 27–37% of the mismatch energy. Re-running the existing 8B adaptation at $r \in \{4, 8, 16, 32, 64\}$ should show a knee near $r \approx 20$ and a collapse at $r \leq 8$. This costs one sweep on an existing pipeline and would convert Theorem 3 from a qualitative obstruction into a quantitative, measured one.

2.7 A sharpening of Theorem 3

Proposition 2.8 (Three distances suffice). *Assume A, B invertible and position-independent, $A^\top R_{\Omega'}(\Delta_i)B = R_{\Omega}(\Delta_i)$ for $\Delta_0 = 0$ and two further distances Δ_1, Δ_2 with $\Delta_2/\Delta_1 \notin \mathbb{Q}$, the frequencies within each table distinct, and the eigenvalue matching induced at Δ_1 and at Δ_2 given by the same permutation. Then $\Omega' = \Omega$ up to sign and permutation.*

Proof. $\Delta_0 = 0$ gives $A^\top B = I$, hence $B = A^{-\top}$ and $SR_{\Omega'}(\Delta_i)S^{-1} = R_{\Omega}(\Delta_i)$ with $S = A^\top$. Similarity preserves spectra, so $\{e^{\pm i\omega'_k \Delta_i}\} = \{e^{\pm i\omega_k \Delta_i}\}$ for $i = 1, 2$. Under the common-matching hypothesis, for each k there are integers m, n with $\omega'_k - \omega_k = 2\pi m/\Delta_1 = 2\pi n/\Delta_2$, so $m/n = \Delta_1/\Delta_2 \notin \mathbb{Q}$, forcing $m = n = 0$. $\square$

This tightens the manuscript’s hypothesis (“an open interval”, or integer positions up to the 2π alias) to a finite, protocol-realisable set of distances, which matters because finite-distribution protocols are what the mature-scale experiments actually measure.

2.8 The ceiling of Part II

Everything above bounds **emulation**: making the Ω' model compute the Ω model’s function. LoRA fine-tuning does **re-adaptation**: it minimises the LM loss *in the Ω' coordinate system*, and nothing forces it to pass through the Ω solution. The paper’s own co-adaptation result — that the two self-consistent pairs are tied at PPL ≈ 7 while both mismatched cells collapse — says precisely that a good Ω' solution exists; the obstruction only says you cannot get there by *copying* the Ω one.

So no linear-algebraic argument can produce a rank floor for the quantity the experiments measure. Breaking this ceiling requires bounding $\min_{\text{rank } \Delta W \leq r} \mathcal{L}(W + \Delta W)$ from below, which needs a loss-landscape tool, not a spectral one: a rank-restricted Fisher/NTK capacity bound (the achievable loss decrease is controlled by the trace of the Fisher information restricted to the rank- r

tangent manifold), or a statistical-dimension argument for the class of functions a rank- r -perturbed head can express on the training distribution.

There is, however, an honest intermediate claim that the manuscript can make now, and that Theorem 2.2 supports exactly:

On a subspace of codimension $\leq 2r$ the adapted head is bit-identical to the naive transplant. Any capability that requires modifying the head’s positional response on more than $2r$ independent directions is therefore unreachable at rank r , regardless of the training objective.

Remark 2.9 (An explanation the paper currently lacks). The 8B LoRA results show in-window PPL *degrading* ($6.82 \rightarrow 10.07$ at 8K) while long-range PPL improves by $8\times$ ($991 \rightarrow 128$ at 32K). Theorem 2.2 plus Table 4 predicts exactly this asymmetry: the long-range response lives in the slow-channel subspace, which has effective rank 2 and is therefore repaired first and almost for free; the in-window response lives in the fast channels, which are mutually near-orthogonal and consume rank one-for-one. A rank- r budget spent greedily therefore buys long range before it buys in-window fidelity. This is a mechanism, testable by the rank sweep above.

3 Part III: non-uniform separation priors

3.1 App. A.1 is prior-agnostic

Proposition 3.1 (Cross-Gram for an arbitrary separation prior). *Let μ be any distribution on relative separations with characteristic function $\psi_\mu(t) = A(t) + iB(t)$, $A(t) = \mathbb{E}_\mu \cos(t\Delta)$, $B(t) = \mathbb{E}_\mu \sin(t\Delta)$. Then with $d = \omega - \nu$, $s = \omega + \nu$ and $x_\omega(\Delta) = [\cos \omega\Delta \quad \sin \omega\Delta]$,*

$$H_{\omega\nu} = \mathbb{E}_\mu[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} A(d) + A(s) & B(s) - B(d) \\ B(s) + B(d) & A(d) - A(s) \end{bmatrix}, \quad S_\omega = H_{\omega\omega}. \quad (16)$$

App. A.1’s formula is the special case $\mu = \text{Unif}[0, L]$, for which $A(t) = \sin(tL)/(tL)$ and $B(t) = (1 - \cos tL)/(tL)$, i.e. the paper’s a and $b_\star$.

Proof. Product-to-sum on each of the four entries and linearity of $\mathbb{E}_\mu$. $\square$

Two immediate consequences worth stating in the manuscript. (i) *The spectral budget identity is prior-independent.* Theorem 1 ($r_2(R) = 2K/[1 + (K - 1)\bar{c}]$) uses only $R_{ii} = I_2$ and $R_{ji} = R_{ij}^\top$, both of which hold for (16) whenever $S_\omega \succ 0$. Nothing about the uniform prior enters. (ii) *The whole of App. A.1 becomes a one-line substitution $(a, b_\star) \mapsto (\text{Re } \psi_\mu, \text{Im } \psi_\mu)$, which is a cheap and strengthening edit.*

3.2 The collapse coefficient for an arbitrary prior

Theorem 3.2 (General low-frequency collapse rate). *Let μ have finite sixth moment $m_j = \mathbb{E}_\mu[\Delta^j]$ and support on at least four points, let $G_0 = \begin{pmatrix} 1 & m_1 \\ m_1 & m_2 \end{pmatrix} \succ 0$ be the Gram of $V_0 = \text{span}\{1, \Delta\}$ in $L^2(\mu)$, let $\Pi_0^\perp$ be the $L^2(\mu)$ -orthogonal projection off V_0 , and let M be the 2×2 Gram of the pair $(-\frac{1}{2}\Pi_0^\perp \Delta^2, -\frac{1}{6}\Pi_0^\perp \Delta^3)$. Then as $\epsilon = \max\{\omega, \nu\}\sqrt{m_2} \rightarrow 0$,*

$$2 - \|Q_{\omega\nu}\|_F^2 = \Lambda(\mu) (\omega^2 - \nu^2)^2 + O(\epsilon^6), \quad \boxed{\Lambda(\mu) = \text{tr}(G_0^{-1}M)}. \quad (17)$$

For $\mu = \text{Unif}[0, L]$ this evaluates to $\Lambda = \frac{19}{12600}L^4$, recovering App. A.1’s $2 - \|Q\|_F^2 = \frac{19}{12600}(x^2 - y^2)^2$ with $x = \omega L$, $y = \nu L$.

Proof. Parameterise by $\theta = \omega^2$ and take the (non-orthonormal) basis $u_1(\theta) = \cos(\omega\Delta)$, $u_2(\theta) = \sin(\omega\Delta)/\omega$ of V_θ , which is smooth in θ with $V_0 = \text{span}\{1, \Delta\}$ and $\dot{u}_1 = -\Delta^2/2$, $\dot{u}_2 = -\Delta^3/6$ at $\theta = 0$. The sum of squared sines of the principal angles between two nearby 2-planes $V_\theta, V_{\theta'}$ equals $(\theta - \theta')^2 \|\Pi_0^\perp \dot{U} G_0^{-1/2}\|_F^2 + O(|\theta|^3)$, where $\dot{U} = [\dot{u}_1, \dot{u}_2]$ and $G_0^{-1/2}$ whitens the base-point basis; the whitening is exactly the $S^{-1/2}$ in $Q = S_\omega^{-1/2} H S_\nu^{-1/2}$, and $2 - \|Q\|_F^2 = \sum_i \sin^2 \varphi_i$. Expanding the norm gives $\text{tr}(G_0^{-1} M)$, and $(\theta - \theta')^2 = (\omega^2 - \nu^2)^2$. $\square$

Verification. Numerically, $\Lambda(\text{Unif}[0, 1]) = 1.5079516 \times 10^{-3}$ against $19/12600 = 1.5079365 \times 10^{-3}$ (agreement to 1×10^{-5} relative), and the leading term matches the exact $2 - \|Q\|_F^2$ to 0.06% at $(\omega, \nu) = (0.05, 0.10)$ — the manuscript’s own quoted check (1.00058). Theorem 3.2 therefore reproduces App. A.1 exactly and extends it to every prior with finite sixth moment.

The paper’s headline figure survives. Computing the block-whitened r_2 of the 23 pairs with $\omega L \leq 1$ at $L=4096$, $b=5 \times 10^5$, $K=64$ under $\mu_\alpha \propto \Delta^{-\alpha}$ on $[1, L]$ gives $r_2 = 2.000$ at $\alpha = 0, 1, 2$ alike ($\bar{c} \geq 0.9998$ in every case). **The $r_2 = 2.00$ collapse claim is prior-robust and can be stated as such.** This is worth one sentence in the manuscript, because it is the claim a reviewer is most likely to attack on prior-realism grounds.

3.3 What changes: fast-end redundancy

Proposition 3.3 (Fast-pair redundancy is the prior’s characteristic function). *If $\psi_\mu(\omega + \nu) \rightarrow 0$ (both channels fast relative to the prior’s resolution), then $S_\omega, S_\nu \rightarrow \frac{1}{2}I$ and (16) gives $Q_{\omega\nu} \rightarrow \begin{pmatrix} A(d) & -B(d) \\ B(d) & A(d) \end{pmatrix}$, hence*

$$c_{\omega\nu} \longrightarrow |\psi_\mu(\omega - \nu)|^2. \quad (18)$$

(Numerically exact to 5 digits at $\alpha = 0$; 8% at $\alpha = 1$ and 12% at $\alpha = 2$, where the hypothesis $\psi_\mu(s) \approx 0$ is itself violated.)

(18) is the whole story of Task 3. Packing channels at the fast end buys independence only as fast as $|\psi_\mu|$ decays, and $|\psi_\mu|$ decays at a rate set by the tail index.

Theorem 3.4 (Tail-index limits). *Let μ_α have density $\propto \Delta^{-\alpha}$ on $[1, L]$.*

- (i) $\alpha < 1$: $|\psi_{\mu_\alpha}(t)| \sim (1 - \alpha)\Gamma(1 - \alpha) (tL)^{\alpha-1}$ for $tL \gg 1$. Power decay with exponent $1 - \alpha$; $\alpha = 0$ recovers $|\sin(tL)/(tL)| \sim (tL)^{-1}$.
- (ii) $\alpha \rightarrow 1$ (Zipf): for $1/L \ll t \ll 1$, $A(t) = \frac{\text{Ci}(tL) - \text{Ci}(t)}{\log L} = \frac{\log(1/t) - \gamma}{\log L} + O\left(\frac{1}{tL \log L}\right)$: **logarithmic decay**. Redundancy between two fast channels separated by d is $c \approx [\log(1/d)/\log L]^2$, which is $O(1)$ for all $d \ll 1$.
- (iii) $1 < \alpha < 3$, $\alpha \neq 2$: $\psi_{\mu_\alpha}(t) = 1 - c_\alpha |t|^{\alpha-1} + O(t^2)$ as $t \rightarrow 0$ (the standard non-analytic Pareto term; at $\alpha = 2$ it is $|t| \log(1/|t|)$). $|\psi| \rightarrow 1$, **i.e. redundancy $\rightarrow 1$, at rate $|t|^{\alpha-1}$** .
- (iv) $\alpha > 3$: $\psi(t) = 1 - \frac{1}{2}m_2 t^2 + O(t^4)$ with $m_2 = O(1)$. The prior has a finite intrinsic scale $\sigma_\mu = \sqrt{m_2} \ll L$, and the resolution threshold $\omega L \approx x_0$ of §1 becomes $\omega \sigma_\mu \approx x_0$: a strictly larger fraction of the table is in the collapsed regime.

So the heavy tail acts on the geometry in *two opposite directions*: it pushes the collapse threshold to higher ω (which argues for EVQ-COSH more strongly), and it flattens $|\psi_\mu|$ at the fast end so that packing channels there stops buying independence (which argues against). The net effect must be computed.

Table 5: Block-whitened Rényi-2 effective rank r_2 of the full $K=64$ table at $L=4096$, $b=5 \times 10^5$, under $\mu_\alpha \propto \Delta^{-\alpha}$ on $[1, L]$. $\tau = 0$ is the geometric midpoint table. Bold marks the maximiser.

α	$\tau=0$	1	2	3	3.5	4	5	6	8	12
0.0	7.84	9.81	17.27	34.23	47.27	62.56	93.17	113.65	127.31	127.81
0.5	7.52	9.34	16.09	30.81	41.89	54.63	80.62	99.59	115.40	119.07
1.0	5.43	6.32	8.91	12.58	14.48	16.22	19.05	21.01	23.12	24.14
1.5	3.75	4.04	4.64	5.17	5.35	5.47	5.61	5.66	5.64	5.44
2.0	3.22	3.37	3.63	3.79	3.82	3.82	3.80	3.76	3.65	3.46
2.5	2.92	3.02	3.18	3.25	3.25	3.24	3.20	3.15	3.05	2.89
3.0	2.57	2.64	2.76	2.83	2.84	2.84	2.82	2.79	2.72	2.61

3.4 The overload turnover

Proposition 3.5 (High-frequency overload). *Under μ_α , the map $\tau \mapsto r_2$ is (a) monotone increasing and saturating at $2K$ for $\alpha \leq 1$, so the static criterion alone selects no finite τ ; and (b) unimodal with an interior maximum for $\alpha \geq 1.5$, at $\tau^* \approx 6, 4, 3.5, 3.5$ for $\alpha = 1.5, 2, 2.5, 3$. The mechanism is Prop. 3.3 plus Theorem 3.4(iii): for $1 < \alpha < 3$ the fast-end pairwise redundancy approaches 1 like $|\omega - \nu|^{\alpha-1}$ instead of decaying like $|\omega - \nu|^{-1}$, so concentrating channels at the fast end past τ^* costs more in mutual redundancy than it gains in resolution.*

Verdict on the overload question. High-frequency overload is **real** and appears for $\alpha \gtrsim 1.5$; it is **invisible** under the uniform prior used in App. A.1, where r_2 is monotone in τ and the criterion is degenerate. But it **does not reverse the ordering**: $\text{EVQ-COSH} \geq \text{Geo}$ at every α tested, with the margin shrinking from $8.0 \times$ ($\alpha=0, \tau=4$) to $2.9 \times$ ($\alpha=1$) to $1.17 \times$ ($\alpha=2$) to $1.11 \times$ ($\alpha=2.5$). **The mitigation of slow-frequency collapse survives under a Zipf prior; the magnitude of the benefit does not.** Two honest statements follow for the manuscript: the static advantage quoted at $\alpha=0$ is an upper bound on what a realistic prior would show, and the deployed $\tau \in [0.35, 4]$ happens to lie below or near $\tau^*(\alpha)$ for every $\alpha \geq 1.5$, so the deployed operating points are not in the overload region.

Lemma 3.6 (Prior-side ceiling). *Let $\tilde{X} \in \mathbb{R}^{n \times 2K}$ be the $\sqrt{\mu}$ -weighted, block-whitened feature matrix and $M = \tilde{X} \tilde{X}^\top$. Since $\text{tr}(M^2) \geq \sum_i M_{ii}^2$,*

$$r_2(R) \leq \frac{(\sum_i M_{ii})^2}{\sum_i M_{ii}^2}. \quad (19)$$

*Numerically at $L=4096, K=64$ this ceiling is ≈ 4000 at $\alpha=0$ (inactive), 71–93 at $\alpha=1$, and 6.8–8.4 at $\alpha=2$, against measured r_2 of 3.3–3.8: **under a heavy tail the prior, not the table, becomes the binding constraint on positional rank.** Note that the naive participation ratio $1/\sum_i \mu_i^2$ (which is 2.5 at $\alpha=2$) is not a valid ceiling — whitening reweights the rare long distances — and should not be quoted as one.*

3.5 Agreement with Part I, and the limitation that remains

The two halves of this note reach the operating point by disjoint routes, and it is worth checking whether they agree. They do — but only once the measurement of §1.9 is used in place of the intuition.

The apparent conflict. Read naively, (8) says $\tau_* \propto d_{\text{eff}}/\sqrt{L_{\text{eff}}^J}$, so concentrated attention (L_{eff}^J small) should push τ *up* without bound, while Prop. 3.5 says a concentrated separation prior makes the redundancy-optimal τ turn over and settle at $O(1)$. These are opposite prescriptions.

The conflict dissolves. Prop. 1.10 shows the naive reading has the sign wrong: because $g_i(j) = O(r_{ij})$, attention locality *raises* L_{eff}^J , and the measured $L_{\text{eff}}^J/L_{\text{train}} \approx 4$ pushes τ *down* by a factor ≈ 0.5 (Cor. 1.11). Prop. 3.5 also pushes τ down: at $\alpha \geq 1.5$ the deployed $\tau \in [0.35, 4]$ already sits at or below $\tau^*(\alpha) \in [3.5, 6]$, and the static margin at $\tau = 4$ has shrunk by $7\times$ relative to $\alpha=0$. **Two independent routes — a measured softmax-Jacobian normalisation and a static Gram computation under a realistic prior — both say the deployed τ is on the high side, by roughly a factor two.** This is evidence, not proof: the routes are not independent of the same modelling choices, and the $0.5\times$ training arm that would decide it does not exist.

The limitation that survives.

Proposition 3.7 (Diagnosis). *The A.10 utility $U(\tau) = d_{\text{eff}}\kappa_{\text{att}} \int \rho_\tau(\phi)q(\omega(\phi)L_{\text{rng}})d\phi$ is **diagonal in the channel index**: a ρ -weighted sum of per-channel phase variances, with no cross-channel term. The cross-channel term is $c_{\omega\nu}$, and by Prop. 3.3 it equals $|\psi_\mu(\omega - \nu)|^2$, negligible under μ_0 (decay $(|\omega - \nu|L)^{-1}$) but $O(1)$ under μ_α , $\alpha \geq 1$. Substituting L_{eff}^J for L repairs the normalisation of the diagonal term and leaves the omitted off-diagonal term untouched. The operating-point model therefore cannot see the overload of Prop. 3.5 at all: its agreement with the geometry in §3.5 is a coincidence of sign, not a derivation.*

This is the theoretical ceiling of Part I. The constant-free rule (8) is only trustworthy while the separation prior is close enough to uniform that the off-diagonal term is negligible — and the very regime where the L_{eff}^J substitution changes the answer is the regime where that condition fails.

What would break it. The surrogate $K_{\text{app}} = \alpha\delta + \beta \min(\phi, \psi)$ was fitted (App. A.9) to the collision kernel of the *uniform* prior. The min kernel is the Green function of $-\partial_\phi^2$ — a short-range, local operator — and that locality is what turns the Euler equation into an ODE and the answer into a cosh. Under μ_1 , Prop. 3.3 and Theorem 3.4(ii) give

$$c_{\omega\nu} \approx \left[\frac{\log(1/|\omega - \nu|)}{\log L} \right]^2, \quad (20)$$

a **logarithmically singular, long-range kernel**. The corresponding variational problem is not a Green’s-function ODE but a **logarithmic-energy / equilibrium-measure problem** in the sense of potential theory. The relevant tools are the Frostman equilibrium measure and Riesz-energy minimisation, and the qualitative prediction they make is concrete and different: the minimiser of a logarithmic energy on an interval is the *arcsine* law $\rho(\phi) \propto [\phi(1 - \phi)]^{-1/2}$, which piles mass at *both* ends of the band. The heavy-tail-optimal allocation should therefore be an arcsine-type density tilted toward the fast end, not a monotone cosh — a two-sided table with a depleted middle. That is fully specified and costs nothing to test: evaluate r_2 under μ_α for an arcsine-tilted table and compare against ρ_τ at its own $\tau^*(\alpha)$.

4 Summary

What now stands on firmer ground. $c(\text{II})$ is not a protocol constant: its full (L, b) dependence is $c = \sqrt{g(\varphi_*)/g_{\text{max}}}$ with $g(\varphi) = \varphi(1 - \varphi)(2 - \varphi)$, $g_{\text{max}} = 2\sqrt{3}/9$, $\varphi_* = \log(L/2.074)/\log b$, and

this reproduces every deployed τ in the paper to 5–9% with no fitting. $c = 1$ is the model’s ceiling, attained on $b = (L/x_0)^{(3+\sqrt{3})/2}$. The repository’s own κ_{att} audit supplies the missing input: $L_{\text{eff}}^J/L_{\text{train}} \approx 4$ on both self-consistent 50M cells, giving a contractive fixed point at $\tau \approx 1.43$, i.e. $0.51\times$ the deployed rule. Theorem 3 upgrades from a qualitative obstruction to an explicit frozen subspace of codimension $\leq 2r$, giving $r \geq d_{\text{rot}}/2$ for exact single-distance repair (the deployed $r=64$ sits exactly there) and $r \geq (1 - \eta) \sum_k s_k/8$ in general, with $\sum_k s_k \rightarrow \sigma_\mu^2 KW_2^2(\Omega, \Omega')$ in the small-shift limit. App. A.1 becomes prior-agnostic by a one-line substitution, the $r_2=2.00$ figure is prior-robust, and the $19/12600$ coefficient is recovered as the uniform case of $\Lambda(\mu) = \text{tr}(G_0^{-1}M)$.

What is refuted or must be softened. $c = 1$ does not follow from the model (F1–F4), and once the measured L_{eff}^J is used the rule overshoots by $\approx 2\times$. The $b=256$ fixed-support control sits on the formula’s degeneracy locus and its deployed $\tau=4$ is $1.7\times$ the model’s optimum. The rank analysis does *not* separate allocation from support — it separates slow-only from full-band. High-frequency overload is real for $\alpha \geq 1.5$ and erases most of the static EVQ-COSH margin, though it never reverses the ordering.

The two ceilings, stated plainly. (1) The obstruction bounds *emulation*, not *re-adaptation*, and closing that gap needs a rank-restricted Fisher/NTK capacity bound rather than linear algebra. (2) The operating-point utility is diagonal in the channel index; under a heavy-tailed prior the omitted off-diagonal term dominates, and the L_{eff}^J substitution repairs only the normalisation. The fact that Parts I and III both end up pushing τ down by $\approx 2\times$ is therefore corroboration by two routes, not a derivation. Closing *that* gap requires re-deriving the surrogate under the realistic prior, where the kernel is logarithmic rather than Green, and the natural tool is equilibrium-measure / logarithmic-potential theory — with the concrete prediction that the optimum becomes arcsine-type rather than cosh.

Three cheap experiments this note implies. (i) Add a $\tau = 0.5\times$ arm to the existing staged multiplier study — the self-consistent fixed point built from the repository’s own κ_{att} measurement predicts the optimum there, and no reported sweep goes below $0.75\times$. (ii) Sweep the 8B LoRA rank over $\{4, 8, 16, 32, 64\}$ and look for the predicted knee at $r \approx 20$. (iii) Recompute the App. A.1 diagnostics under μ_α with α measured from the training corpus’s attention statistics, and report the α -conditioned r_2 margin instead of the $\alpha=0$ one.

Reproduction. Every number in this note is produced by `verify_three_completions.py` (pure NumPy/SciPy, ~ 30 s CPU), which also runs the five lemma verifications of §2 to machine precision.